Additional links and prototypes

Everything for Round 1 in one document: the live viewer, the capture app, the technical abstract, the repository and the demonstration video. All web links need no login and were checked before this document was written.

1 · Live web viewer (free roaming)

What	Link
Landing film and entry point	`https://ricco.codes/ply/`
Classroom twin, first-person walkthrough	`https://ricco.codes/ply/walk.html?scene=https%3A%2F%2Fply-api.ricco-ply.workers.dev%2Fv1%2Fscenes%2Fhq-classroom-1%2Fscene.json`
Classroom semantic model	`https://ricco.codes/ply/model.html?scene=https%3A%2F%2Fply-api.ricco-ply.workers.dev%2Fv1%2Fscenes%2Fhq-classroom-1%2Fscene.json&module=/ply/models/classroom2.js`
Office LiDAR point cloud	`https://ricco.codes/ply/lidar.html?ply=https%3A%2F%2Fply-api.ricco-ply.workers.dev%2Fv1%2Fscenes%2Flidar-office-1%2Foffice.ply`
Capture app (records and uploads)	`https://ricco.codes/ply/capture.html`
Capture procedure, rendered	`https://ricco.codes/ply/guide.html`

Controls. Click the walkthrough to capture the mouse, then W A S D to walk and the mouse to look. `Esc` releases the pointer. **Views** opens saved shots, **Plan** shows the floor plan, **Measured** overlays the derived shell. Search any of the 194 places to route to it.

Browsers. Chrome or Edge on desktop for the full walkthrough; Safari on macOS and iOS loads the model and the capture pages. AR navigation needs WebXR, which today means Android Chrome. Nothing needs an install or an account.

2 · Technical abstract

`Ply_Technical_Abstract_Round1.pdf`, five pages, in this folder. It covers system architecture, SOP construction, pipeline design and the technical approach, with the measured numbers for both capture paths.

3 · Code repository, setup and sample data

What	Where
Repository	`https://github.com/Ricco-Mallick/ply`
Setup	`README.md` and `pipeline/README.md` (Python 3.12, CUDA GPU, `python -m plypipe run <input> --out <dir>`)
Sample data	`test/` with a short capture and the command to reconstruct it
Capture procedure	`docs/SOP.md` (also `/ply/guide.html`)
Processing policy	`docs/PROCESSING.md`

4 · Demonstration video

Three to five minutes: the problem, the capture walk, the processing run, the web walkthrough with search and routing, and the AR overlay. LINK ADDED BEFORE UPLOAD

5 · A five-minute tour for reviewers

1. Open `ricco.codes/ply` and scroll through the film, or press **Skip to the end** to reach the entry card.
2. **Start a capture** walks the capture flow; **Explore the demo space** opens the twin.
3. In the walkthrough, click once to take the mouse, then W A S D to walk. **Plan** gives the floor plan, **Dollhouse** the cutaway.
4. Search a place by name and follow the route. The line bends around desks, not through them.
5. On Android Chrome, the **AR** button places the same route on the real floor with two taps to align.

6 · Reproducing a room

```
python -m plypipe run <folder of clips> --out work/my-room --preset standard
# writes scene.json, shell.json, places.json, scene.spz, scene_lite.spz, report.json
# then open walk.html?scene=<url of scene.json>
```

Settings are planned from the capture, so there is nothing to tune between rooms. `docs/PROCESSING.md` explains what is measured and why, including the post-run checklist.

STATUS BEFORE UPLOAD

- Repository is private today. Either make it public, or put a source archive (`ply-src.zip`) in this folder and point the row above at it.
- `test/` does not exist yet. A short sample capture plus a run command has to be added for the repository to satisfy the brief.
- The demonstration video is not recorded. It is the only Round 1 deliverable that has no artefact yet.

THIRD-PARTY COMPONENTS

three.js (MIT), @sparkjsdev/spark (MIT), React (MIT), Vite and TypeScript (MIT), Tailwind (MIT), SuperSplat (MIT, PlayCanvas), gsplat (Apache-2.0), COLMAP and pycolmap (BSD-3-Clause), PyTorch (BSD-3-Clause), OpenCV (Apache-2.0), Hugging Face Transformers (Apache-2.0), @playcanvas/splat-transform (MIT). Full notices with copyright lines are in `NOTICE`.